Chuanben Li

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Education

University College London *Sep 2023 - Present*
BSc Physics (reading)

- **2nd year and 1st year**: both expecting a **1st Class Honours** , equivalent of at least **3.7** in the **U.S. system**, or a minimum of **1.5** in the **German system**) (Oxford official grade reference manual: The UK Honours Degree System for Undergraduates)

Research Experience

Undergraduate Physics Group Project *London, UK*
University College London *Jan 2026 - Mar 2026*

- ***Supervisor*: Prof. Ferruccio Renzoni (*Full Professor at UCL*)**
- **Topic: Magnetic Field Synthesis**
- Implemented computational models of planar inductive coil arrays to achieve low-frequency magnetic field focusing and steering, using Biot–Savart law integration and matrix-based optimisation (Direct Matrix Algorithm and pseudo-inverse method).

UCL Brian Duff and AMOPP Undergraduate Summer Research Studentship *Remote*
University College London *Jul 2025 - Present*

- ***Advisor*: Prof. Carla Figueira de Morisson Faria (*Full Professor at UCL*)**
- **Topic: Investigation of electron dynamics in Kramers-Henneberger (KH) atom via laser fields**
- Constructed numerical solutions of the time-dependent Schrödinger equation to investigate IR and XUV pump-probe schemes for controlling coherent superpositions of KH eigenstates.
- Characterized coherent superpositions by computing autocorrelation functions and analyzing phase-space signatures, generating time-dependent wavefunction probability density plots for different KH eigenstates.
- Verified the Hamiltonian transformation from length gauge to velocity gauge and subsequently to the Kramers–Henneberger (KH) frame, comparing standard total-field and strong-field-only approaches for consistency with theoretical formulations.
- Designed and implemented quantum gates by tuning laser field parameters; computed autocorrelation functions, identified turning points and global minima, calculated state populations, and visualized laser fields and envelopes; determined phase coefficients at the conclusion of each laser application.

Summer Research Experience Program in Molecular Science (SRPMS) *Shanghai, CN*
NYU Shanghai *Jun 2025 - Aug 2025*

- ***Advisor*: Prof. Xiang Sun (*Assistant Professor at NYU Shanghai*)**
- **Topic: Simulating population trajectories of real molecules via the Fewest Switches Surface Hopping (FSSH) Algorithm**
- Applied Fewest Switches Surface Hopping (FSSH) algorithm in **Python** to simulate time-dependent population trajectories of adiabatic and diabatic states sampled from the Wigner distribution for standard Tully models including Simple Avoided Crossing (SAC), Dual Avoided Crossing (DAC), and Extended Coupling with Reflection (ECR).
- Plotted adiabatic potential energy curves and non-adiabatic coupling (NAC) strength as a function of position for the Tully models to support analysis of the dynamics simulations.
- Employed **Gaussian 16, ORCA, and OpenMolcas** to perform ground state geometry frequency optimisation, Wigner sampling, TDDFT, CASSCF, Spin-Orbit Coupling (SOC), constructed input files and successfully tested the workflow and functionalities of **SHARC** software to perform *ab initio* **FSSH** nonadiabatic dynamics population simulations of photodissociation of gas-phase phenyl azide (Ph-N_3).

Ocean Carbon Capture via Electrochemical Reactions *Shanghai, CN*
Independent Researcher *Jun 2024 - Sep 2024*

- Investigated marine carbon sequestration mechanisms through independent experimentation
- Designed and executed a seawater carbon mineralization study by adding NaOH to coastal seawater samples (Lingang, Shanghai), successfully precipitating 0.7 g of Ca^{2+}/Mg^{2+}minerals
- Established an electrochemical proof-of-concept reaction circuit with an anion exchange membrane (AEM), observing carbonate precipitation through reduction of water at the cathode, and validating the feasibility for scalable ocean alkalinity enhancement (OAE).

Wave Energy Converter *Shanghai, CN*
Independent Researcher *Jun 2024 - Sep 2024*

- Investigated the feasibility of using unanchored wave energy converters (WECs) for providing energy for global-scale ocean carbon capture.
- Conducted literature review on WEC mechanisms, identifying technical difficulties for achieving unanchored and robust stability against extreme marine conditions.
- Developed a novel prototype of a floating WEC:
  - Designed magnetic ball-inductor system in buoyant housing to test unanchored energy generation.
  - Optimized coil winding density and swing amplitude to achieve measurable current output (lighting a bulb).
  - Field-tested at coast of Lingang, Shanghai, and analyzed failure modes due to low wave amplitudes.

Extended Project Qualification (EPQ) *Thetford, UK*
Thetford Grammar School *Oct 2021 - Sep 2022*

- ***Advisor*: Mrs. Tracy Granger (*Head of Science at Thetford Grammar School*)**
- **Topic: Investigating the oil-removing mechanism of wheat bran in local dishwashing**
- Researched the mechanism and experimentally validated the low-cost and biodegradable oil-removal efficacy of wheat bran compared to chemical dish detergents, which was observed during the field trip in Alxa League, Inner Mongolia, China.
- Demonstrated the physical mechanism of oil absorption by cellulose fibers, highlighting its potential for reducing water pollution through hydrophobic functional group substitutions

Academic Enrichment

DESNEZ – Carbon Capture and Hydrogen Projects (USI 2026) *Remote*
Project Member — Carbon Capture & Storage (CCS) Infrastructure *Feb 2026 – Mar 2026*
University College London

- Contributed to an interdisciplinary analysis of UK CCS cluster development (HyNet, East Coast, Acorn, Viking), focusing on infrastructure coupling between CO_2 capture, transport, and geological storage.
- Evaluated economic risk-allocation mechanisms for early-stage CCS deployment and assessed engineering challenges in repurposing oil and gas infrastructure, including integrity, compatibility, and leakage risk.
- Examined legal and liability frameworks for long-term CO_2 storage in depleted reservoirs and co-authored a focused report contributing to a consolidated annual submission intended for policymakers.

Oxford Quantum Club *Remote*
University of Oxford *Nov 2021 - May 2022*

- Organized by Prof. Alexander Lvovsky, A-Level students participated in weekly online workshops led by Oxford penultimate year students covering undergraduate-level of quantum mechanics:

Leadership Experience

ASDAN Global Volunteer Program Inner Mongolia Wildlife and Environment Conservation *Alxa League, Inner Mongolia Autonomous Region, CN*
Field trip *Jul 2021*

- Participated in a field-based camp near the Helan Mountains, observing the impact of climate change on

local desertification and drying mountain channels, motivating my focus on carbon capture.

- Conducted site visits, local interviews, and academic research on topics of desertification, water conservation, traditional land use, and organic farming.
- Engaged in hands-on environmental work, including drought-resistant saxaul planting and high-altitude hiking in arid terrain.
- Collaborated in teams to synthesize findings and co-deliver a final research presentation and written report

Caojiadu Street Library, Jingan District, Shanghai: "Wonderful Library Night" Activity Certificate of Voluntary Service *Shanghai, CN* *Aug 2018*
Volunteer

- Helped coordinate and deliver a nighttime reading event at the library, engaging primary school children with storytelling and interactive activities.

Shanghai Rendu Ocean NPO Development Centre: "Love my life source" Beach Cleaning Project *Shanghai, CN* *Jun 2018*
Volunteer

- Took initiative during coastal fieldwork in Nanhuizui, Lingang (Shanghai), helping coordinate the removal and classification of marine debris as part of a broader environmental conservation effort.

Honors and Certificates

The Rhodes Scholarship (China Constituency) Semifinalist *Oct 2025*

Brian Duff and AMOPP Undergraduate Summer Research Studentship *Jun 2025*

BPhO (British Physics Olympiad) Round 1 Top Gold (top 50) *Nov 2022*

BPhO Physics Computational Challenge (Gold Distinction – top 9, actually ranked 5th) *Mar 2022 - Aug 2022*

BPhO Gold Prize (top 150) *Nov 2021*

NSDA Certificate of Achievement for 3rd Place in Public Forum Debate Novice Division Suzhou Regional Championship *Suzhou, China* *May 2019*
NSDA China

- Topic: "*On balance*, refugee restrictions in developed countries are permissible."
- As the second speaker of our two-people team, I led argument development, cross-fire preparation, and evidence collection. We were the first team from our new school to win a regional tournament.
- Ranked 7th overall entering elimination rounds and placed 3rd in the final.
- One of 500 national qualifiers selected from over 10,000 participants across 18 regional tournaments for the NSDA National Final Championship in China.

NSDA Certificate of Achievement for 7th Outstanding Speaker in Public Forum Debate Novice Division Suzhou Regional Championship *Suzhou, China* *May 2019*
NSDA China

- Due to my outstanding performance in the public forum debate tournament, I was awarded the 7th Outstanding Speaker among hundreds of participants in the Suzhou Regional.

Computing skills

Programming Languages: Python (machine learning), C++

Softwares: TensorFlow, SHARC, OpenMolcas, ORCA, Gaussian 16, Gromacs, Open Babel, Conda, Mathematica, Solidworks CAD (CSWP Certification), CuPy.

Operating Systems and Environments: Linux cloud-based virtual environments, HPC clusters (Slurm/SGE), Windows CMD